Sujit Mandava

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Education

Northwestern University
Master of Science in Artificial Intelligence *Sept 2025– Dec 2026*

Indian Institute of Technology Palakkad
Bachelor of Technology in Computer Science and Engineering *Oct 2020– Jul 2024*

Experience

Applied Scientist Intern, Relativity **June 2026– August 2026**

- Built an agentic system to surface supporting evidence for legal claims across a 47K-document OCR'd corpus (hybrid RAG, agentic retrieval, LLM-judge ranking), reaching 0.72 recall at $17/claim vs. 0.90 at $400 for exhaustive LLM review: 80% of ceiling recall at 4% of the cost.
- Raised retrieval from **0.3 recall@1000** to **0.68 recall@1000** with agentic retrieval over a claim-anchored pipeline (dense retrieval, LLM claim decomposition, phrase match, RRF fusion), and selected a **770M-parameter local verifier** (MiniCheck, AUC 0.780) over a frontier-model judge (0.736) at **∼400x lower cost with zero document egress**.
- Closed the human-in-the-loop with **GEPA**-optimized ranking prompts on reviewer rerun; SME validation on a claim surfaced **13 supporting citations missed in the original filing**, a **62% increase**.

Software Engineer, hGrid (Mahathi Software) **Dec 2024– Aug 2025**

- Owned **design and implementation of the full-stack architecture** for three production products, **CareTab**, **SunPay**, and **Docuracy**, making architecture decisions independently in a **zero-legacy** environment and backing all three on **Amazon RDS**.
- Built **deployment, observability, and monitoring pipelines** from scratch with **Docker**, **AWS**, **nginx**, **gunicorn**, **Grafana**, and **Loki**, cutting incident detection from user reports to dashboard alerts across all three products.
- Implemented **asynchronous task orchestration** for CareTab using **Celery**, **Django**, and **django-celery-beat**, moving scheduled and long-running jobs off the request path to keep API latency stable under load.

Projects

Wren: Retrieval-Augmented Caregiver Assistant for Dementia Care

- Built **Ask Wren**, a full-stack RAG pipeline giving caregivers a conversational chat interface over patient data.
- Designed **context retrieval** pulling recent sessions, transcripts, check-ins, and notes into a grounded prompt for **Azure OpenAI GPT-4o**.
- Implemented JWT auth with rotating refresh tokens and **Argon2** hashing, plus an idempotent demo data seed script.

Cellar: Personalized Wine Discovery and Ranking Platform

- Designed a **comparison-based ranking system** where users place bottles via binary search comparisons instead of typing scores—position in the list *is* the score—with bucket bands, Bayesian-shrunk community averages, and session-resumable placement state.
- Built a **118k-wine catalogue** on Postgres with full-text search (GIN + trigram indexes, 8ms lookups), a grape-variety type-inference pipeline covering 708 varieties, and pagination that cut a 354MB endpoint to bounded responses.
- Shipped **Claude-vision label scanning**: camera capture, structured extraction of producer/vintage/region/grape, trigram-matched catalogue lookup with user confirmation, and a fallback contribute flow—plus a predicted-score engine (taste-match-weighted regression shrunk to content priors) for cold-start recommendations.

Nietzsche Persona Chatbot (GPT-2, FAISS, sentence-transformers)

- Built a pipeline that pulls Nietzsche's works from **Project Gutenberg**, segments them into a training corpus, and **fine-tuned GPT-2** with configurable hyperparameters, checkpoint management, and perplexity-based evaluation.
- Grounded generation in retrieved passages via **FAISS** vector search and sentence-transformer embeddings, with source attribution on every response.

Leadership

Captain & Assistant Coach, IIT Palakkad Basketball Club **Aug 2023– May 2024**

- Founded and coached IIT Palakkad's **first women's basketball team**, mentoring players and coordinating strategy to achieve a **quarterfinals finish** in their inaugural tournament.
- Scaled the club from **8 to 65+ active members** by introducing structured practice systems, governance rules, and accountability mechanisms to support sustained growth.

Technical Skills

Languages: Python, SQL, TypeScript/JavaScript, C++
ML & LLMs: PyTorch, Hugging Face Transformers, sentence-transformers, scikit-learn, FAISS, DSPy/GEPA
Methods: RAG, Agentic Retrieval, Fine-Tuning, LLM Evaluation, Prompt Optimization
Backend & Infra: FastAPI, Django, Celery, SQLAlchemy, PostgreSQL (Amazon RDS), MCP, Docker, nginx, AWS, Azure OpenAI, GCP, Grafana, Loki